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B.Tech. Degree VI Semester Regular/Supplementary Examination June 2023

ME 19-205-0605 CAD/CAM
(2019 Scheme)

Time: 3 Hours

Maximum Marks: 60

Course Outcomes

On successful completion of the course, the students will be able to:

- CO1: Understand the basics of Computer Aided Design (CAD), geometric modelling, automation, flow line analysis, and line balancing.
- CO2: Understand the structure and special design features of CNC machines. Learn manual CNC programming and computer-aided programming technique like APT.
- CO3: Understand the basic configurations, components, applications and the programming of robots.
- CO4: Learn the fundamentals of the advanced concepts in automation and manufacturing like DNC, CIM, FMS, and CAE and the basic concepts of AI and expert systems for manufacturing automation. Learn the basic concepts and applications of Rapid Prototyping.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate, L6 – Create

PO – Programme Outcome

PART A(Answer **ALL** questions)

(8 × 3 = 24)

	Marks	BL	CO	PO
I. (a) Explain: (i) orthographic projection (ii) isometric projection (iii) axonometric projection with illustrations.	3	L2	1	1,2,5
(b) Define and explain the terms with appropriate mathematical expressions: (i) manufacturing lead time (ii) production capacity (iii) production rate (iv) cycle time (v) availability (vi) work in progress.	3	L2	1	1,2,5
Describe the axes' nomenclature of a CNC Turning Centre and CNC Turning Centre with an illustration.	3	L3	2	1,5
(d) What are zero offsets? Explain how zero offset is determined in a CNC lathe.	3	L2	2	1,2,3
(e) Explain the difference between point-to-point and contour type motion systems in robots	3	L2	3	1,2,4
(f) Explain any three examples of end effectors used in robot with neat figures.	3	L2	3	1,2,4
(g) Describe the limitations of rapid prototyping methods.	3	L1	4	1,4,5
(h) List out and explain different levels of automation with illustrations.	3	L1	4	1,2

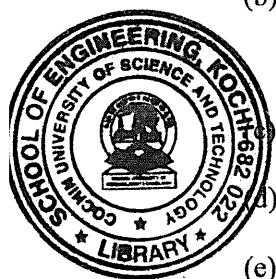
PART B

(4 × 12 = 48)

- II. (a) What are transformation matrices? Give transformation matrices for: (i) Translation (ii) Rotation (iii) Reflection in homogeneous coordinates in 2D. 6 L1 1 1,2,5
- (b) The coordinates of the vertices of a rectangle are (0, 0), (2, 0), (2, 2), and (0, 2). Construct a matrix with homogeneous coordinates of the rectangle and rotate it by 45 degrees about the centre (1, 1). Use appropriate transformation matrices for operation. 6 L3 1 1,2,5

OR

(P.T.O.)



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- III. (a) (i) What are the different plant layouts? Explain with illustrations. Marks 2 BL L1 CO 1 PO 1,4,5
 (ii) What is the line balancing problem? What is balance delay? 2
 (b) In a factory 480 units are to be made in an 8 h (480 min) shift. The activity times and precedence requirements are shown in the table given below. Assuming 100% efficiency, solve the line-balancing problem using the largest candidate rule and find the balance delay. 8 L3 1 1,2,4,5

Task	Task Time (Seconds)	Precedence
A	15	-
B	23	A
C	17	B
D	42	B
E	15	B
F	37	C
G	5	D, E
H	12	F, G
I	34	H
J	27	H
K	18	I, J
L	7	K

- IV. (a) Explain the classification of CNC machines based on control loop used with illustrations. 4 L1 2 1,2,5
 (b) A CNC program for milling a profile is given below. Draw the cutter path and the profile of the job. The thickness of the plate is 5 mm. An HSS endmill, $\phi 20$ mm is used for milling the profile. 8 L3 2 1,2,5

$d=20$ mm;

T1- end mill $\phi 20$ mm

%1234

N1 G54 G90 G94

N2 T1 M06

N3 M03 S330 M08

N4 G00 X-15 Y-15 Z5 (5 mm above the job surface)

N5 G01 Z-10 F100 (move to the required depth)

N6 X-10

N7 Y80

N8 G02 X30 Y110 R40

N9 G01 X74.14

N10 X130 Y54.14

N11 Y-10

N12 X-15

N13 G00 Z100

N14 G28

N15 M02

OR

- V. (a) Describe Recirculating Ball Screws used in CNC machines. How are Ball Screws pre-loaded? How does preloading affect the accuracy of Ball Screws? 6 L2, L3, L5 2 1,2,3,5
 (b) (i) What is machine zero? Illustrate machine zero for CNC Turning Centres and Machining Centres. 6 L3 2 1,2,5
 (ii) What is referencing (Homing) in CNC Machines?

(Continued)

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		Marks	BL	CO	PO
VI.	(a) Explain the structure of a typical robot with neat figure.	6	L2	3	1,2,3
	(b) Explain following kinematic configurations, applications, and its work envelope with neat figure:	6	L2	3	1,2,3
	(i) Cartesian manipulator				
	(ii) Spherical manipulator.				
OR					
VII.	Make a conceptual design of a simple pick and place robot which pick material from one conveyor belt and place it in another conveyor belt. Explain the configuration of the system and the components with the help of neat diagrams.	12	L6	3	1,2,3,4
VIII.	(a) Explain stereolithography with the help of neat figure.	6	L2	4	1,2
	(b) Explain the organizational structure for CIM. Which functional groups come under CIM?	6	L3	4	1,4
OR					
IX.	(a) What are the steps involved in a typical material addition rapid prototyping?	6	L2	4	1,2
	(b) Explain FMS with illustration. Differentiate between FMC and FMS.	6	L2	4	1,2

Bloom's Taxonomy Levels

L1 = 16.7%, L2 = 39.2%, L3 = 32.5%, L5 = 1.6%, L6 = 10%.

